Esther Cheng

Education

Duke University – Durham, NC August 2022 – May 2026
Bachelor of Arts, Computer Science & Psychology (GPA: 3.9)

Relevant Coursework: *Data Structures & Algorithms, Computer Architecture, Operating Systems, Distributed Systems, Advanced Algorithms, Discrete Math, Methods in User-Centered Research*

Work Experience & Organizational Leadership

Deutsche Bank – Cary, NC
Software Engineer, RiskFinder Calculation Services July 2026 – Present
- Streamlined post-release testing with a UI-driven smoke test runner, saving 2 engineering hours monthly and enabling non-technical users to validate releases independently

Software Engineer Intern, Global Receipts Platform June 2025 – August 2025
- Engineered an ESG analytics microservice, enabling document-level sentiment analysis, cross-document comparison, and competitor benchmarking
- Optimized NLP and web-scraping pipelines, reducing document pre-processing runtime by 90%

Duke University Innovation Co-Lab – Durham, NC August 2024 – May 2026
Software Developer
- Engineered a transcription application with speaker diarization and live transcript editing, leveraging open-source speech models to streamline transcript review and reduce manual correction time
- Developed an end-to-end attendance tracking system integrating a Raspberry Pi card reader with a Next.js tablet UI, MongoDB persistence, authentication, and REST APIs for automated student identification and attendance logging
- Integrated a real-time attendance analytics dashboard with existing class-management workflows, enabling administrative overrides and centralized monitoring while increasing staff productivity by 25%

Duke University Computer Science Department – Durham, NC January 2024 – January 2026
Undergraduate Teaching Assistant, ECE/CS 250: Computer Architecture
- Led weekly recitations and office hours for 20+ students, reinforcing concepts in C, MIPS Assembly, and digital logic
- Provided ongoing student support via EDStem, debugging and clarifying architecture and systems-level coursework

Projects

Pickleball Matchmaker | *Next.js, TypeScript, React, PostgreSQL* October 2026
- Cut round-generation and scoring latency by 65% (5.8s to 1.6s), using bulk PostgreSQL multi-row inserts and simulation tuning, to eliminate 20+ sequential database round-trips and achieve sub-500ms UI updates
- Built player profile and membership systems featuring dynamic skill progression charts and role-based visibility controls, using PostgreSQL schemas and granular privacy toggles, enabling persistent career records while safeguarding player data
- Developed full-stack tournament brackets and live open-play queues, using Next.js 15, React Server Actions, and PostgreSQL, to enable automated bracket progressions and zero-loss guest account syncing

BlueBot | *Python/FastAPI, FastMCP, OpenAI, LiteLLM* December 2025
- Engineered MCP tools for an agentic course discovery service integrated with LiteLLM, enabling real-time catalog search within a Duke-specific chatbot

AI Course Advisor | *Python/Flask, TypeScript, React, MongoDB, Redis, Docker, Figma, HuggingFace, OpenAI* July 2024
- Implemented secure Duke SSO using OAuth and a Flask backend-for-frontend (BFF) architecture
- Integrated Azure-hosted GPT-4o with RAG and vector embeddings for semantic course search
- Optimized MongoDB aggregation pipelines with pre-filtering and re-ranking to improve retrieval relevance
- Improved chatbot reliability by 15% through prompt engineering and iterative model fine-tuning

Skills

Languages: Python, Java, JavaScript, TypeScript, Ruby, C, MIPS Assembly, HTML, CSS, YAML, SQL
Libraries/Frameworks: Next.js, Node.js, Express, React, Flask, FastAPI, FastMCP, NLTK, PyTesseract (OCR), PyMuPDF, Dash, REST API, Spring/Spring Boot, Redis, MongoDB, PostgreSQL, Tailwind, BigQuery, LiteLLM
Tools/Platforms: Git, Bash, Google Cloud Platform, Docker, Kubernetes, OpenShift, Helm, Maven, Swagger, Figma, Jira